Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam A · Form 1: Foundations

Multiple Choice Questions: Select the best answer. (1 point each)

1. A book rests on a table. What is the third-law partner of Earth's gravitational pull on the book?

(A) The book pushing the table

(B) The table pushing the book

(C) Earth pulling the table

(D) The book pulling Earth gravitationally

2. A 10 kg box slides right. A 40 N force acts rightward at 0° below horizontal. $\mu_k = 0.15$; contact is maintained and vertical acceleration is zero. Find kinetic friction magnitude. Use g = 9.8 m/s².

(A) -14.7 N

(B) 29.4 N

(C) 7.35 N

(D) 14.7 N

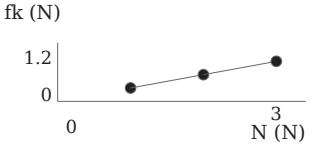
3. The same two surfaces slide against each other. Vary the normal force and measure friction magnitude. N (N) -> fk (N): (1, 0.4); (2, 0.8); (3, 1.2). Which vertical reading does the model predict at zero horizontal coordinate?

(A) 0

(B) 0.1

(C) -0.1

(D) 0.2



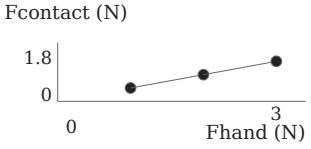
4. Touching 2 kg and 3 kg carts accelerate on a frictionless floor when a hand pushes the 2 kg cart right. Fhand (N) -> Fcontact (N): (1, 0.6); (2, 1.2); (3, 1.8). What is the signed change in the vertical reading from first to last trial?

(A) 0.72

(B) 1.2

(C) 1.68

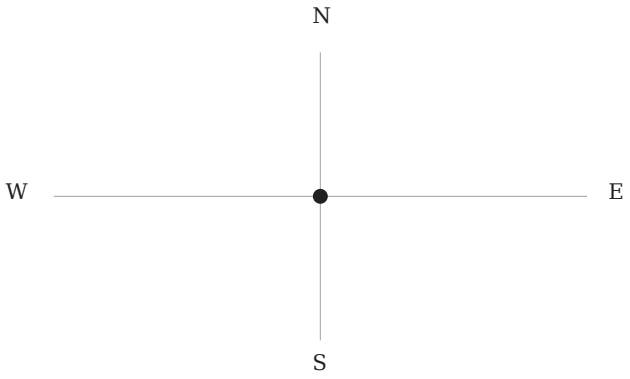
(D) 2.16



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. Two students push a cart of mass 8 kg on a frictionless horizontal floor. A exerts 180 N at 0° north of east. B exerts 260 N at 30° north of west.

a. Draw and label the actual forces acting in the plane of the floor on the dot. Draw whole vectors, not components. **(1 pt)**



b. Show net-force x and y components, then find acceleration magnitude and direction using compass words. **(3 pt)**

6. A 2 kg cart slides right against constant 2 N friction to the left. Fpull (N) -> a (m/s²): (1, -0.5); (2, 0); (3, 0.5).

a. Build the model for cart under tow from physics and predict both slope and intercept. **(2 pt)**

b. Plot the first and last measurements and describe the trend. **(2 pt)**

7. A mass-m cart slides right. Force F acts rightward at θ above horizontal, with kinetic friction coefficient μ. Contact is maintained. Derive horizontal acceleration with zero vertical acceleration. **(3 pt)**